

SECTION C

11 (**long**) The height z of some terrain is given by the equation

$$z = (x + 1)(x + 2y - 2)(3x - 4y - 1)$$

(a) Sketch contours of constant z for the region $-2 < x < 2$ and $-2 < y < 2$. [8]

(b) Show that

$$\begin{aligned}\frac{\partial z}{\partial x} &= 9x^2 + 8y + 4xy - 8y^2 - 8x - 5 \\ \frac{\partial z}{\partial y} &= 2(x + 1)(x - 8y + 3)\end{aligned}\quad [6]$$

(c) Find the stationary points of z and classify them by inspection, without computing second derivatives. [8]

(d) In which direction does the contour of constant z pass through the origin? [4]

(e) Find the rate of change of z at the point $(x, y) = (1, 1)$ in the direction $(-1, -1)$. [4]